Pearson Edexcel International A Level · Pure Mathematics 1

Chapter 7 — Radians
Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P1 第7章 · 弧度制
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

Figure 1 shows a footpath bounded by two concentric arcs AB and CD and two radial edges. The outer arc is twice as long as the inner arc. The two arcs have total length 18 m, and the footpath has area $27\ \text{m}^2$. Let the inner radius be r m and the outer radius be R m. The common angle is θ radians, with $0 < \theta < \pi$.

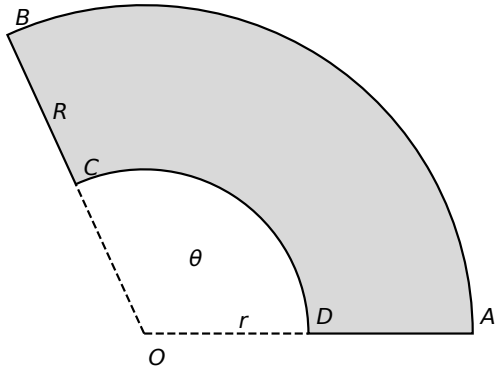


Figure 1

(a) Show that the width $R - r$ of the footpath is 3 m. (3)

(b) Hence find

(i) the inner radius r and the outer radius R, (2)

(ii) the value of θ. (2)

(c) Both radial edges also need edging. Find the total length of edging required. (2)

(Total for Question 1 is 9 marks)

Figure 2 shows a sector OAB of radius 6 cm and angle θ radians, where $0 < \theta < \pi$. Point C lies on OB with $OC = 4$ cm. Triangle OAC is removed from the sector, leaving a panel bounded by the minor arc AB and the straight segments BC and CA. The panel has perimeter 14 cm.

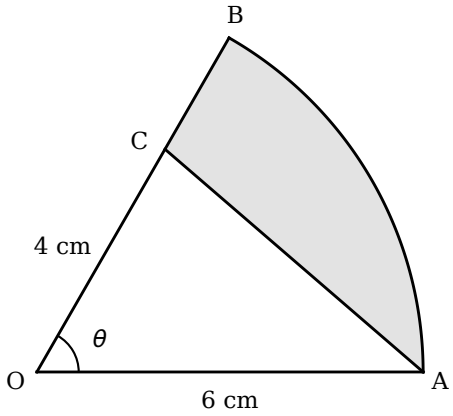


Figure 2

(a) Show that $CA = 12 - 6\theta$ cm, and hence show that $\cos\theta = \dfrac{52 - (12 - 6\theta)^2}{48}$. (3)

(b) A second panel has the same lengths OA, OB, OC but has $\theta = 1.2$; its perimeter is not fixed at 14 cm. Find

(i) the perimeter of the panel to 3 significant figures. (2)

(ii) the area of the panel to 3 significant figures. (2)

(Total for Question 2 is 7 marks)

A flower bed is a sector OAB of radius 8 m and angle 1.6 radians. Its curved fence AB is to be replaced by the straight chord AB, leaving the triangular bed OAB. Figure 3 shows the area that will be lost.

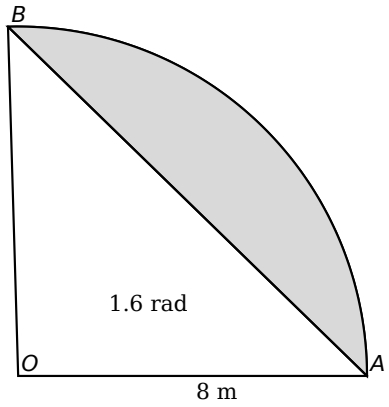


Figure 3

(a) Find the reduction in fence length, to 3 significant figures. (3)

(b) Find the area lost, to 3 significant figures. (3)

(c) The owner claims that more than one third of the original area will be lost. Determine whether the claim is correct. (2)

(Total for Question 3 is 8 marks)

Two straight roads OA and OB meet at an angle of 0.9 radians, with $OA = 5$ km and $OB = 8$ km. A circular road with centre O and radius 5 km joins A to the point C on OB.

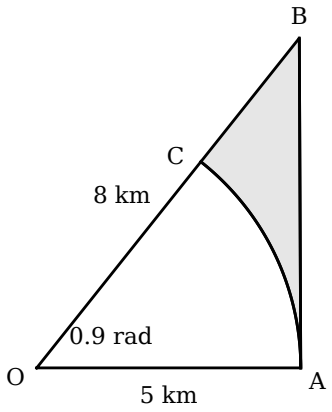


Figure 4

(a) Find, to 3 significant figures, the length of the direct road AB. (2)

(b) The alternative route from A to B follows the minor arc AC and then the straight road CB.

(i) Find the additional distance travelled using this route, to 3 significant figures. (2)

(ii) Find the area bounded by this alternative route and the direct road, to 3 significant figures. (2)

(Total for Question 4 is 6 marks)

Two circular garden beds each have radius 5 m. Their centres O_1, O_2 are 6 m apart. The circles intersect at A, B. The segments AB and O_1O_2 are perpendicular and bisect each other at M. Let $\alpha = \angle AO_1M$. Figure 5 shows the common area of the two beds.

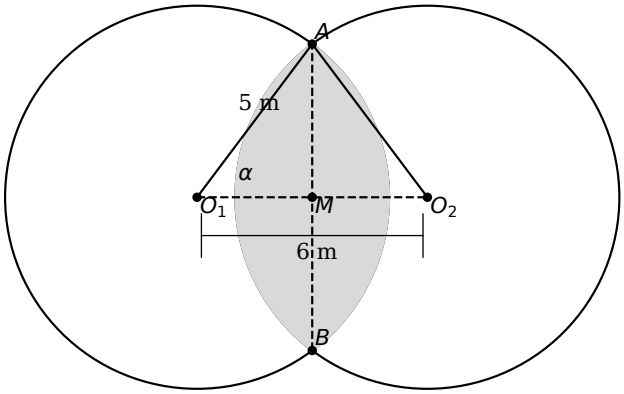


Figure 5

(a) Find AB and give an exact expression for α. (3)

(b) Find the common area, giving your answer in terms of α. (3)

(c) Find the perimeter of the common area, to 3 significant figures. (2)

(Total for Question 5 is 8 marks)

6.

A wheel of radius 0.35 m rolls in a straight line without slipping. A mark P is initially at the point of contact with the ground. The wheel makes 4 complete turns and then turns through a further 1.4 radians.

(a) Find the distance travelled by the centre of the wheel, to 3 significant figures. (3)

(b) For the final position of P, find

(i) its height above the ground, to 3 significant figures. (2)

(ii) the fraction of one full revolution represented by the final 1.4-radian turn. Give your answer to 3 significant figures. (1)

(Total for Question 6 is 6 marks)

A circular plate has centre O and radius 10 cm. Chords AB and AC lie on opposite sides of radius OA, with $\angle AOB = 0.8$ radians and $\angle AOC = 1.1$ radians. Points A, B, C lie on the minor arc from B to C.

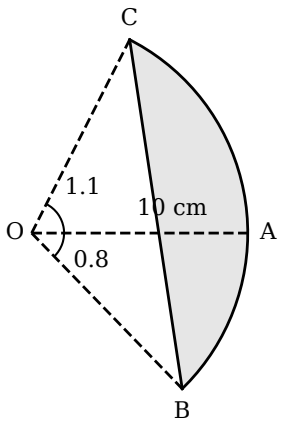


Figure 7

(a) Find the length of this arc from B through A to C. (2)

(b) A thin piece is cut off along chord BC; the piece is bounded by the minor arc BAC and chord BC. Find, to 3 significant figures,

(i) the length of BC. (2)

(ii) the area of the piece removed. (2)

(Total for Question 7 is 6 marks)

Two sector-shaped panels have the same arc length, 12 cm. Their radii are r cm and $(r + 3)$ cm. The sum of their areas is $66\ \text{cm}^2$.

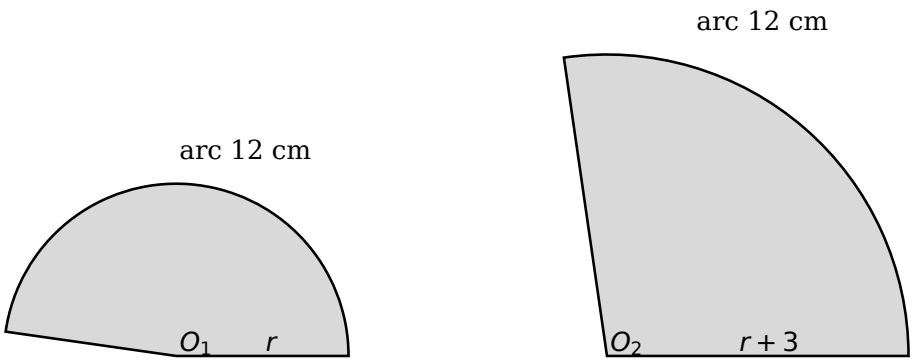


Figure 8

(a) Show that the area of the larger panel exceeds that of the smaller by $18\ \text{cm}^2$, regardless of r. (3)

(b) Find both radii and both angles. (3)

(c) Which panel has the greater perimeter? State by how much. (2)

(Total for Question 8 is 8 marks)

A windscreen wiper sweeps an annular sector of angle 2.2 radians. The distances of the ends of the rubber blade from the pivot are 0.18 m and 0.60 m. Figure 9 shows the swept region.

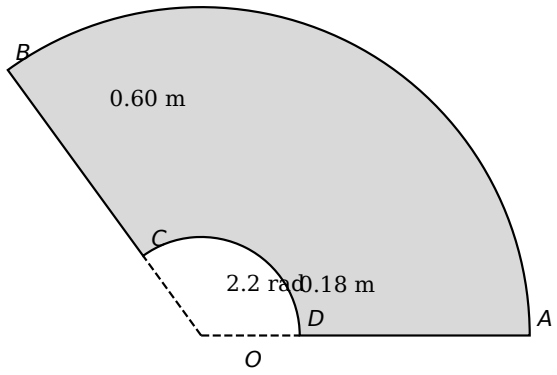


Figure 9

(a) Find the swept area to 3 significant figures. (2)

(b) Find the complete perimeter of the swept region. (3)

(c) The sweep angle may be adjusted to θ, where $0 < \theta < \pi$. The straight-line distance between the two extreme positions of the outer tip must not exceed 1.00 m.

(i) Find the greatest permitted value of θ, to 3 significant figures. (3)

(ii) State, with a reason, whether the original sweep angle meets this requirement. (1)

(Total for Question 9 is 9 marks)

A circular sheet is cut along a chord AB of length 10 cm. The chord subtends an angle of $\pi/3$ at the centre. The larger piece is kept, as shown shaded in Figure 10.

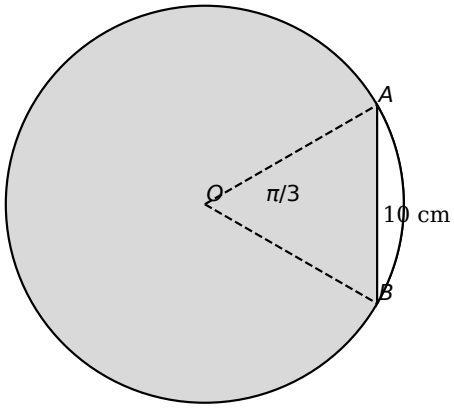


Figure 10

(a) Find the radius of the sheet. (2)

(b) Find the exact area of the piece kept. (4)

(c) Find the exact perimeter of the piece kept. (2)

(Total for Question 10 is 8 marks)

A sign consists of a rectangle of width $2r$ cm and height r cm, surmounted by a semicircle of radius r cm. Its outside perimeter is 20 cm.

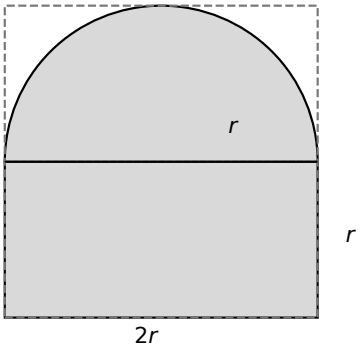


Figure 11

(a) Find r exactly. (2)

(b) Show that the area of the sign is $200/(4+\pi)$ cm^2. (3)

(c) A rectangular board of the same width and full height is used to make the sign. Find the fraction of the board that is cut away. (2)

(Total for Question 11 is 7 marks)

Triangle ABC has $AB = 8$ cm, $AC = 11$ cm and $BC = \sqrt{73}$ cm. A sector of radius 5 cm, centred at A, is removed from the corner. Its arc meets AB, AC at D, E. Figure 12 shows the remaining panel. Let $\alpha = \angle BAC$.

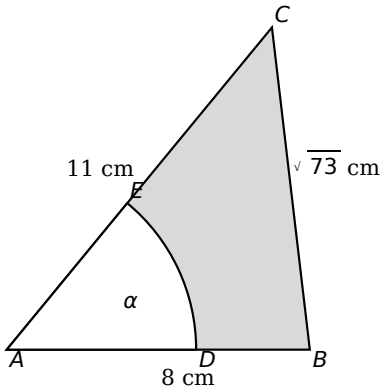


Figure 12

(a) Show that $\cos\alpha = 7/11$, and find $\sin\alpha$ exactly. (3)

(b) Find the remaining area, giving your answer in terms of α. (2)

(c) Find the perimeter of the remaining panel, to 3 significant figures. (3)

(Total for Question 12 is 8 marks)

An arm pivots at O. Starting with length 3 m, its tip moves anticlockwise through an angle θ. The arm is then extended by 2 m without turning. With its new length, it turns through a further angle θ in the same direction. The total tip path is 14 m. Assume $0 < \theta < \pi/2$.

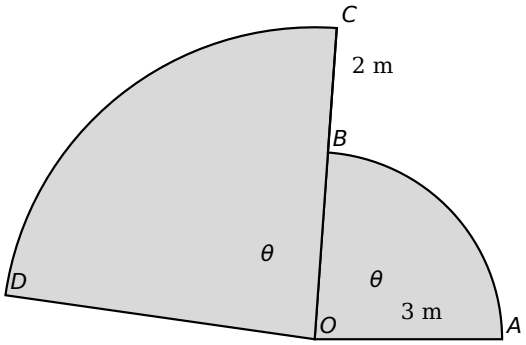


Figure 13

(a) Find θ. (3)

(b) Find the straight-line distance between the initial and final tip positions, to 3 significant figures. (3)

(c) Find the total area swept out during the two turns, excluding the zero-area extension. (2)

(Total for Question 13 is 8 marks)

A C-shaped mask is made from an annulus with outer radius 12 cm and inner radius 6 cm by removing an annular sector of angle ϕ, where $0 < \phi < \pi$. The remaining area is 81π cm^2.

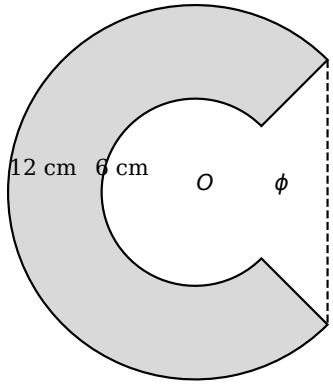


Figure 14

(a) Find ϕ exactly. (3)

(b) Find the exact perimeter of the mask. (3)

(c) A straight support joins the two outer ends across the gap. Find its length exactly. (2)

(Total for Question 14 is 8 marks)

A road of width 4 m consists of two annular sectors joined end to end along a complete radial edge. Each sector has inner radius 8 m, outer radius 12 m and angle $\pi/3$. The first bends left and the second bends right, so that each long side contains one inner arc and one outer arc. Figure 15 shows the plan.

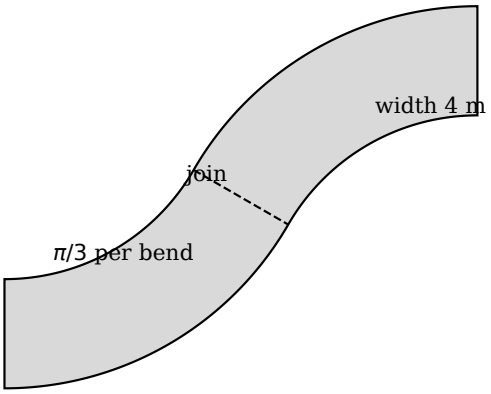


Figure 15

(a) Find the length of each curved long side exactly. (2)

(b) Find the total road area exactly. (2)

(c) Edging is required around the whole outside boundary, including the two open ends. Find the length required. (3)

(Total for Question 15 is 7 marks)

A circular window has radius 13 cm. A straight horizontal bar cuts off a minor segment whose greatest height above the bar is 8 cm. Let α be half the angle subtended by the bar at the centre, as shown in Figure 16.

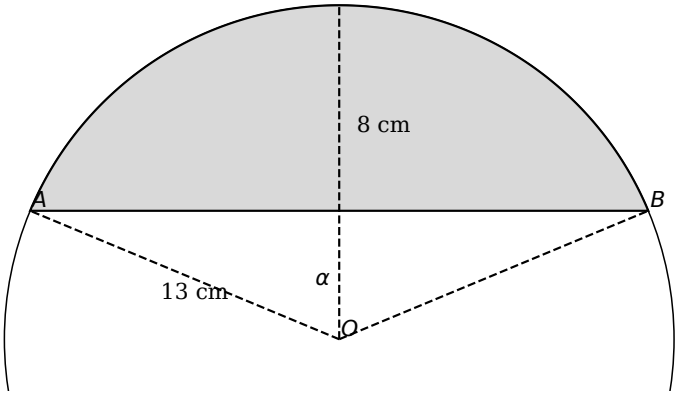


Figure 16

(a) Find the length of the bar and an exact expression for α. (3)

(b) Find the exact area of the segment, in terms of α. (3)

(c) Find the perimeter of the segment to 3 significant figures. (2)

(Total for Question 16 is 8 marks)

A quarter-disc of radius r is divided by the chord joining its two arc endpoints. Figure 17 shows the right-angled triangle next to the centre and the remaining segment.

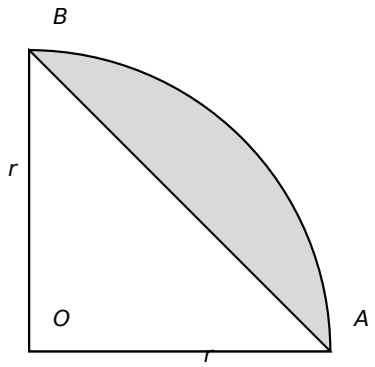


Figure 17

(a) Show that the triangle occupies the fraction $2/\pi$ of the quarter-disc, independently of r. (3)

(b) The triangle has area $18\ \text{cm}^2$. Find the exact segment area. (2)

(c) Use the result of part (a) to explain why no positive value of r makes the triangle occupy 70% of the quarter-disc. (2)

(Total for Question 17 is 7 marks)

A sector has radius r cm and angle θ radians, where $0 < \theta < \pi$. Its perimeter is 30 cm. The length of its arc exceeds the length of its chord by 2 cm.

(a) Show that $r = \dfrac{30}{2 + \theta}$ and hence show that $30\theta - 60\sin(\theta/2) = 4 + 2\theta$. (3)

(b) A trial angle is $\theta = 1.5$. Using the perimeter constraint,

(i) find the arc length minus the chord length, to 3 significant figures. (2)

(ii) state whether this trial angle satisfies the required difference of 2 cm. (1)

(Total for Question 18 is 6 marks)

19.

A clock has a minute hand of length 12 cm. At 12:00 the hand points vertically upwards. The hand moves clockwise at a constant rate.

(a) Find, in radians, the angle through which the hand turns in 18 minutes. Give your answer in terms of π. (2)

(b) Between 12:00 and 12:18, find

(i) the exact distance travelled by the tip, and the magnitude of its straight-line displacement to 3 significant figures. (3)

(ii) the area swept out by the hand, in exact form. (2)

(Total for Question 19 is 7 marks)

Three gates A, B, C lie consecutively on a minor circular arc, with common centre O. The arcs AB and BC have lengths 8 m and 12 m, respectively, and $\angle AOC = 2$ radians. A new straight path is proposed from A to C.

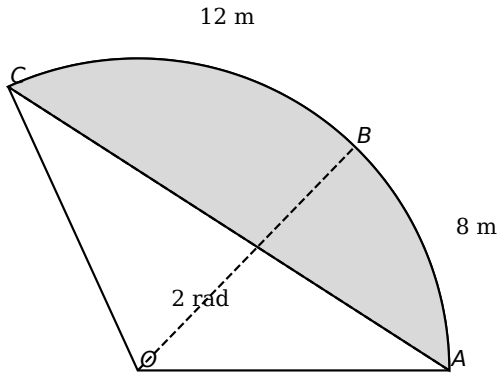


Figure 20

(a) Find the radius and the angles AOB and BOC. (3)

(b) Find the distance saved by using the new path rather than the curved route through B. (2)

(c) Find the area between the new path and the old route, to 3 significant figures. (3)

(Total for Question 20 is 8 marks)

A wedge of angle θ radians is removed from a circular sheet of radius 5 cm, with $0 < \theta < \pi$. The remaining major sector must have area at least 60 cm^2 and perimeter no greater than 35 cm.

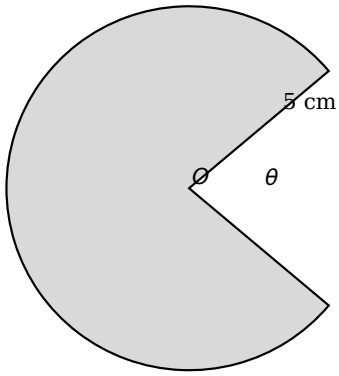


Figure 21

(a) Obtain one inequality in θ from each design requirement. (4)

(b) Give the full set of permitted values of θ. (2)

(c) Find the smallest possible straight-line gap between the two outer ends of the cut, to 3 significant figures. (3)

(Total for Question 21 is 9 marks)

Two concentric circles have centre O and outer radius 17 cm. A chord AB of the outer circle has length 30 cm. The perpendicular from O meets the midpoint of AB and has length equal to the inner radius r. The radii OA, OB cut the inner circle at D, C. Figure 22 shows the annular sector $ABCD$.

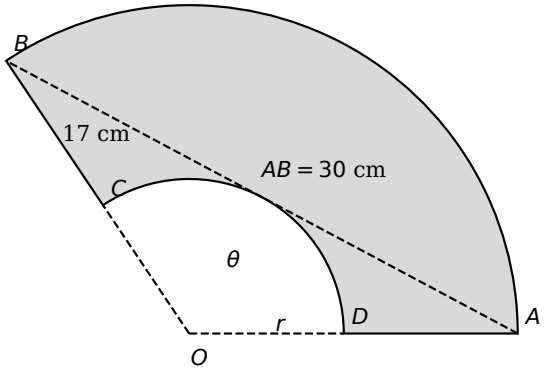


Figure 22

(a) Find r and the common angle θ. (3)

(b) Find the area of the annular sector in terms of θ. (2)

(c) Find the complete perimeter of this annular sector, to 3 significant figures. (3)

(Total for Question 22 is 8 marks)

A wire 18 cm long is bent into a circular arc. Joining its ends to the circle centre forms a sector of area $36\ \text{cm}^2$. The sector is the one bounded by the wire itself, as shown in Figure 23.

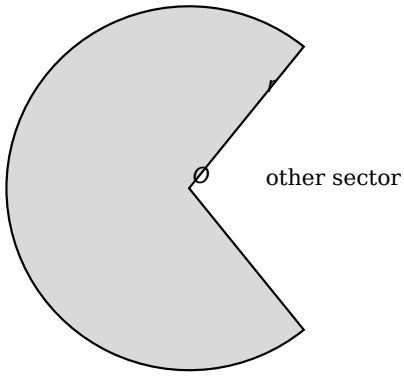


Figure 23

(a) Find the radius and the angle of this sector. (3)

(b) Explain whether this is a minor or major sector, and find its complete perimeter. (3)

(c) Find the exact area of the other sector determined by the same two radii. (2)

(Total for Question 23 is 8 marks)

A sector-shaped garden of radius r m is divided into two smaller sectors by one radial fence. Their angles are θ and 2θ. All the outer boundary and this internal fence use 54 m of fencing. The garden has area 108 m^2 and total angle less than π.

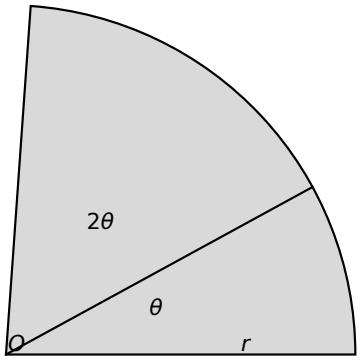


Figure 24

(a) Show that $r^2 - 18r + 72 = 0$. (3)

(b) Find the radius and both smaller angles, explaining which root must be rejected. (3)

(c) Find how much fencing is around the outside boundary only. (2)

(Total for Question 24 is 8 marks)

A rectangle of width 12 cm stands on the diameter of a semicircle of radius 10 cm. Its upper vertices A, B lie on the arc and its base is centred on O, the circle centre. The semicircle outside the rectangle consists of a cap above AB and two side regions. Figure 25 shows these regions.

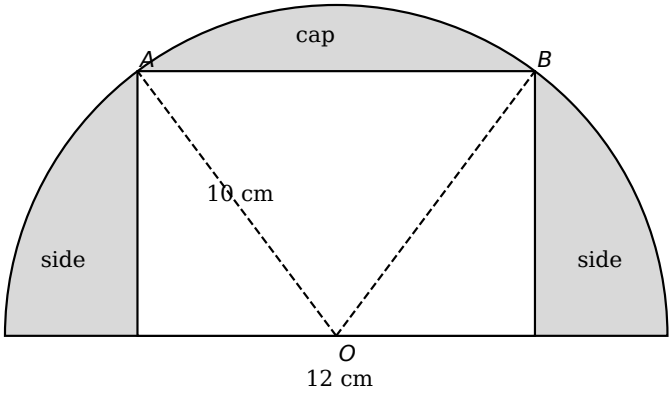


Figure 25

(a) Find the rectangle height, and show that $\angle AOB = 2\arcsin(3/5)$. (3)

(b) Let $\alpha = \arcsin(3/5)$.

(i) Find the area of the cap in terms of α. (3)

(ii) Find the combined area of the two side regions, and hence state which is larger: the cap or the side regions together. Give both areas to 3 significant figures. (3)

(Total for Question 25 is 9 marks)

A sector of radius r cm and angle θ radians has area $54\ \text{cm}^2$. Its arc is 3 times as long as one radius, and $0 < \theta < 2\pi$.

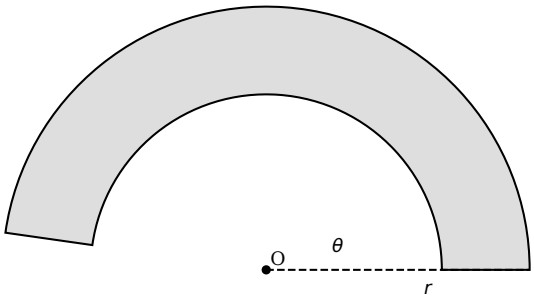


Figure 26

(a) Find r and θ. (3)

(b) A smaller concentric sector with the same angle is removed, leaving an annular sector of area $30\ \text{cm}^2$. Find

(i) the inner radius, (3)

(ii) the complete perimeter of the remaining region. (2)

(Total for Question 26 is 8 marks)

A circular window has radius 10 cm. A horizontal bar of length 12 cm forms a chord. The perpendicular from the centre bisects the chord. The smaller region above the bar is removed.

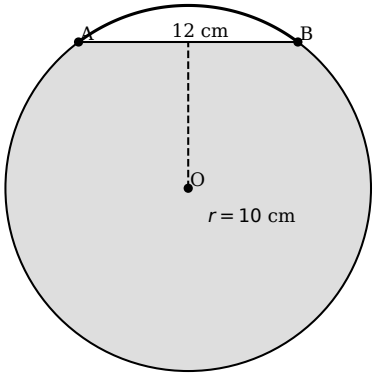


Figure 27

(a) Find the distance from the centre to the bar and the angle subtended by the bar at the centre, in radians to 3 significant figures. (4)

(b) For the remaining larger piece, find

(i) the exact area in terms of $\alpha = \arcsin(3/5)$, (3)

(ii) the complete perimeter in terms of α. (3)

(Total for Question 27 is 10 marks)

A swinging arm of length 2.4 m moves from angle $-\phi$ to angle ϕ relative to the downward vertical, where $0 < \phi < \pi/2$. The tip rises 0.6 m from its lowest position at each end of the swing.

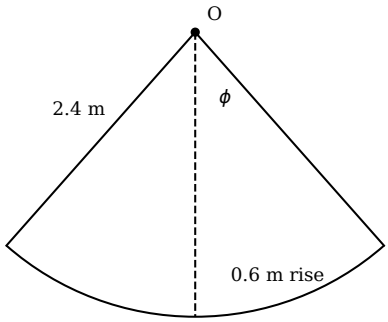


Figure 28

(a) Show that $\cos\phi = 3/4$ and find ϕ to 3 decimal places. (3)

(b) During one complete swing from the left-hand extreme to the right-hand extreme, find, to 3 significant figures,

(i) the distance travelled by the tip, (2)

(ii) the straight-line distance between the two extreme positions. (2)

(Total for Question 28 is 7 marks)

A rectangular sheet has dimensions 18 cm by 10 cm. Two semicircles of radius 5 cm are cut out, one from each short edge, with their diameters on those edges. Their interiors do not overlap.

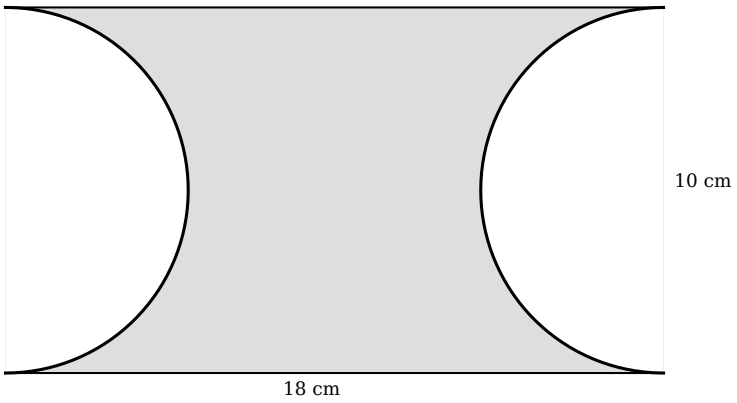


Figure 29

(a) Find the exact area and complete perimeter of the remaining sheet. (4)

(b) Each semicircular cut is replaced by a sector with the same radius and with its arc length reduced by 20%. For this comparison, consider one cut only. Find

(i) the new sector angle, (2)

(ii) the decrease in the area of the cut. (2)

(Total for Question 29 is 8 marks)

A sector-shaped sign has radius r cm and angle θ radians, where $0 < \theta < \pi$. A straight chord joins its arc endpoints. The arc length is 16 cm and the chord length is $8\sqrt{3}$ cm.

(a) Show that $\sin(\theta/2)/(\theta/2) = \sqrt{3}/2$. (3)

(b) A designer proposes $\theta = \pi/3$.

(i) Use the arc length to find the radius for this proposal. (2)

(ii) Determine whether this proposal also gives the required chord length. Give numerical evidence. (3)

(Total for Question 30 is 8 marks)

Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

Question 1

P1-C07-R01 · 9 marks

(a) · 3 marks

$R - r = 3$.

两弧总长 $(R+r)\theta = 18$；面积 $\frac{1}{2}(R-r)(R+r)\theta = 27$。用前一关系代入得 $9(R-r) = 27$。

M1	建立两弧总长关系
M1	把面积写成半径和与差的乘积
A1	推出宽度3

(b) Hence find

(i) · 2 marks

$r = 3$ m, $R = 6$ m.

同角弧长之比等于半径比，R=2r，结合(a)半径差3。

M1	使用弧长比和(a)
A1	两半径

(ii) · 2 marks

$\theta = 2$ radians.

两弧总长9θ=18。

M1	代入两半径求角
A1	弧度值

(c) · 2 marks

24 m.

边界包括两弧18m与两条各3m的直边，总长 $18 + 2 \times 3 = 24$。

M1	计入两条径向直边
A1	完整边界长度及单位

Question 2

P1-C07-R02 · 7 marks

(a) · 3 marks

$CA = 12 - 6\theta$; stated relation.

边界含$BC = 2$、弧$AB = 6\theta$及CA，没有已剪去的OA。因此$CA = 14 - 2 - 6\theta = 12 - 6\theta$。在$OAC$中余弦定理给$CA^2 = 36 + 16 - 48\cos\theta$，整理即得。

M1	正确弧长与周长
M1	余弦定理
A1	关系式

(b) A second panel has the same lengths OA, OB, OC but has $\theta = 1.2$; its perimeter is not fixed at 14 cm. Find

(i) · 2 marks

15.1 cm.

此设计$CA = \sqrt{52 - 48\cos 1.2} = 5.882\ldots$；周长仅为$BC + \text{arc } AB + CA = 2 + 7.2 + 5.882\ldots = 15.082\ldots$，即$15.1$ cm。

M1	以指定角度重新计算斜边
A1	周长

(ii) · 2 marks

10.4 cm^2.

面板面积为扇形减去OAC：$18(1.2) - 12\sin 1.2 = 10.4155\ldots$。

M1	扇形减三角形
A1	面积

Question 3

P1-C07-R03 · 8 marks

(a) · 3 marks

$12.8 - 16\sin 0.8 \approx 1.32$ m.

原弧长为$8\times 1.6=12.8$；将等腰三角形二等分，弦长为$16\sin 0.8$。相减才是节省长度。

M1	弧长
M1	半角求弦长
A1	差值及精度

(b) · 3 marks

$32(1.6 - \sin 1.6) \approx 19.2$ m^2.

失去的是弓形，面积为扇形 $\tfrac{1}{2}r^2\theta$ 减去三角形 $\tfrac{1}{2}r^2\sin\theta$。

M1	扇形减三角形
A1	正确精确式
A1	面积及精度

(c) · 2 marks

Correct: the lost fraction is $(1.6 - \sin 1.6)/1.6 \approx 0.375$.

用损失面积除以原扇形面积51.2，约37.5%，大于$33\tfrac{1}{3}\%$；不能把长度缩短的比例当作面积比例。

M1	用原面积作分母
A1	有数值依据的判断

Question 4

P1-C07-R04 · 6 marks

(a) · 2 marks

6.27 km.

余弦定理$AB = \sqrt{25 + 64 - 80\cos 0.9} = 6.266673\ldots$。

M1	余弦定理并使用弧度
A1	长度

(b) The alternative route from A to B follows the minor arc AC and then the straight road CB.

(i) · 2 marks

1.23 km.

弧AC长$5(0.9) = 4.5$，$CB = 3$，路线共7.5；比直达多$7.5 - 6.266673\ldots = 1.23$。

M1	弧长加直线并比较
A1	路程差

(ii) · 2 marks

4.42 km^2.

该区域为三角形OAB减扇形OAC，面积$20\sin 0.9 - \frac{1}{2}(25)(0.9) = 4.4165\ldots$。

M1	三角形与扇形面积差
A1	面积

Question 5

P1-C07-R05 · 8 marks

(a) · 3 marks

$AB = 8$ m, $\alpha = \arccos(3/5)$.

公共弦在两圆心中点处，三角形O₁MA直角，O₁M=3、斜边5，得AM=4。弦长2AM=8，cosα=3/5。

M1	二等分对称图形及勾股
A1	弦长
A1	弧度角表达式

(b) · 3 marks

$50\alpha - 24$ m^2.

重叠区由两个相同弓形组成；每个扇形角2α，面积25α，减去底8、高3的三角形面积12，再乘2。

M1	识别两个弓形
M1	扇形与三角形的面积
A1	合并正确

(c) · 2 marks

$20\alpha \approx 18.5$ m.

边界是两条角2α、半径5的圆弧；公共弦AB在内部，不应计入。

M1	只计两圆弧
A1	周长及精度

Question 6

P1-C07-R06 · 6 marks

(a) · 3 marks

9.29 m.

总转角$8\pi + 1.4$弧度；无滑动距离$r\theta = 0.35(8\pi + 1.4) = 9.286\ldots$。

M1	完整转数化为弧度
M1	使用无滑动弧长
A1	距离

(b) For the final position of P, find

(i) · 2 marks

0.291 m.

完整转数不改变标记的相对位置。相对最低点转过1.4，高度$0.35(1 - \cos 1.4) = 0.2905\ldots$。

M1	以半径竖直投影求高度
A1	数值

(ii) · 1 marks

0.223.

一周是2π弧度，比例$1.4/(2\pi) = 0.2228\ldots$。

B1	弧度与整周比例

Question 7

P1-C07-R07 · 6 marks

(a) · 2 marks

19 cm.

总圆心角$0.8+1.1=1.9$弧度，弧长$10(1.9)=19$。

M1	正确相加圆心角
A1	弧长

(b) A thin piece is cut off along chord BC; the piece is bounded by the minor arc BAC and chord BC. Find, to 3 significant figures,

(i) · 2 marks

16.3 cm.

等腰三角形BOC平分后$BC=20\sin(1.9/2)=16.268\ldots$。

M1	弦长与半角
A1	数值

(ii) · 2 marks

$47.7\ \text{cm}^2$.

弓形面积为扇形减三角形：$50(1.9-\sin 1.9)=47.685\ldots$。

M1	扇形减三角形
A1	面积

Question 8

P1-C07-R08 · 8 marks

(a) · 3 marks

Difference $= \frac{1}{2}(12)(r+3) - \frac{1}{2}(12)r = 18$.

由弧长l=rθ，扇形面积可写½rl；同弧长下两面积差为½l乘半径差。

M1	推出A=rl/2
M1	代入同弧长与半径差
A1	完整证明

(b) · 3 marks

Radii $4, 7$ cm; angles $3, 12/7$ radians, respectively.

两面积分别6r和6(r+3)，和66给r=4；各角为12除以对应半径。

M1	用面积和求r
A1	两半径
A1	两弧度角

(c) · 2 marks

The larger-radius panel, by 6 cm.

两弧相等，周长差仅来自两条半径，总差2×3=6。

B1	正确选择
B1	长度差与依据

Question 9

P1-C07-R09 · 9 marks

(a) · 2 marks

$0.360\ \text{m}^2$.

有效区域是两扇形之差：½(0.60²−0.18²)×2.2=0.36036。

M1	环形扇形面积
A1	数值及单位

(b) · 3 marks

$2.556\ \text{m}$.

两弧长合为(0.60+0.18)×2.2=1.716，两条直边各0.42，总长2.556。

M1	两弧
M1	计入两端的刀片长
A1	正确周长

(c) The sweep angle may be adjusted to θ, where $0 < \theta < \pi$. The straight-line distance between the two extreme positions of the outer tip must not exceed $1.00\ \text{m}$.

(i) · 3 marks

$\theta = 2\arcsin(5/6) \approx 1.97$ radians.

外端弦长1.20sin(θ/2)在此范围随θ增加；令弦长为1求边界角。

M1	正确弦长条件
M1	使用指定范围求边界
A1	最大角

(ii) · 1 marks

No, since $2.2 > 2\arcsin(5/6)$.

比较原角与(i)未舍入边界，原角更大，弦长超限。

B1	有数值比较支持的判断

Question 10

P1-C07-R10 · 8 marks

(a) · 2 marks

10 cm.

半弦5、半角π/6，5=r sinπ/6=r/2。

M1	二等分弦所对三角形
A1	半径

(b) · 4 marks

$250\pi/3 + 25\sqrt{3}$ cm^2.

小弓形面积为 $50(\pi/3 - \sin\pi/3) = 50\pi/3 - 25\sqrt{3}$；从整圆100π中扣除。

M1	计算小弓形
A1	小弓形精确式
M1	从完整圆扣除
A1	所留面积

(c) · 2 marks

$50\pi/3 + 10$ cm.

保留的是大弧，其角为2π−π/3=5π/3；边界还含弦10，故答案如此。

M1	选择大弧角
A1	曲边加切口的完整周长

Question 11

P1-C07-R11 · 7 marks

(a) · 2 marks

$r = 20/(4+\pi)$ cm.

外边包含底边2r、两侧共2r和半圆弧πr；半圆直径是内部接缝，不计入。

M1	正确外周长式
A1	精确r

(b) · 3 marks

$A = 2r^2 + \pi r^2/2 = \tfrac{1}{2}(4+\pi)r^2 = 200/(4+\pi)$.

将矩形与半圆面积相加，先保留因式再代r，约去一个(4+π)，避免过早小数化。

M1	面积组合
M1	提取公共因式并代r
A1	推出目标

(c) · 2 marks

$(4-\pi)/8$.

整板宽2r、高2r，面积$4r^2$；切去部分为上半个$2r\times r$矩形减半圆，即$(2-\pi/2)r^2$；除以$4r^2$。

M1	整板与切除面积正确
A1	约分后的比例

Question 12

P1-C07-R12 · 8 marks

(a) · 3 marks

$\cos\alpha = 7/11$, $\sin\alpha = 6\sqrt{2}/11$.

余弦定理给(64+121−73)/(2×8×11)=7/11。三角形内角正弦为正，用直角三角形或平方关系得6√2/11。

M1	余弦定理
A1	正确余弦
A1	正弦的精确正值

(b) · 2 marks

$24\sqrt{2} - 25\alpha/2$ cm^2.

三角形面积½×8×11×sinα=24√2，扣除半径5、角α的扇形。

M1	三角形与扇形面积
A1	正确相减结果

(c) · 3 marks

$9 + \sqrt{73} + 5\arccos(7/11) \approx 21.9$ cm.

剩余两直边为8−5和11−5，另有BC及圆弧5α，总和9+√73+5α；α须用弧度。

M1	剩余直边及BC
M1	加入新圆弧而非两半径
A1	周长及精度

Question 13

P1-C07-R13 · 8 marks

(a) · 3 marks

$\theta = 1.5$ radians.

路径由长度3θ的弧、长度2的直线、长度5θ的弧组成，故8θ+2=14。

M1	区分转动弧与伸长直线
A1	正确方程
A1	角度

(b) · 3 marks

$\sqrt{34 - 30\cos 3} \approx 7.98$ m.

两极端位置离O为3和5，夹角为2θ=3；余弦定理得距离平方9+25−30cos3。

M1	使用总角2θ而非θ
M1	余弦定理
A1	距离及精度

(c) · 2 marks

25.5 m^2.

两转动扇形占不同方向角区间，内部不重叠；面积½×9×1.5+½×25×1.5=25.5。

M1	两半径对应两扇形
A1	总面积及单位

Question 14

P1-C07-R14 · 8 marks

(a) · 3 marks

$\phi = \pi/2$.

完整圆环面积108π；剩余角为2π−φ，所以54(2π−φ)=81π，解得φ=π/2。

M1	完整圆环或剩余角面积
M1	建立方程
A1	精确角

(b) · 3 marks

$27\pi + 12$ cm.

两圆弧的剩余角都是3π/2，弧长共18×3π/2=27π，两条切口长度各6。

M1	使用剩余角而非切除角
M1	加两条切口
A1	完整周长

(c) · 2 marks

$12\sqrt{2}$ cm.

支撑是半径12、所对小角π/2的弦，长2×12×sinπ/4=12√2；不是缺口外弧长。

M1	弦长公式或勾股
A1	精确长度

Question 15

P1-C07-R15 · 7 marks

(a) · 2 marks

$20\pi/3$ m.

每条长边各含$8\times\pi/3$和$12\times\pi/3$，不是把两外弧归到同一边。

M1	每侧配一内弧一外弧
A1	两边共同长度

(b) · 2 marks

$80\pi/3$ m^2.

一段环形扇形面积$\tfrac{1}{2}(144-64)\pi/3=40\pi/3$，共两段，不重叠。

M1	一段环形扇形面积
A1	合计面积

(c) · 3 marks

$40\pi/3+8$ m.

两条长边共$40\pi/3$，只有首末两条宽4的径向边属于外界；中间接缝不计入。

M1	两曲边合计
M1	区分两端与内部接缝
A1	完整边界长度

Question 16

P1-C07-R16 · 8 marks

(a) · 3 marks

Bar length 24 cm; $\alpha = \arccos(5/13)$.

圆心到横杆距离为13−8=5；半弦长 $\sqrt{(169-25)}=12$，整杆24；cosα=5/13。

M1	由弓高找圆心至弦距离
M1	勾股求整弦
A1	长度与角表达式

(b) · 3 marks

$169\alpha - 60$ cm^2.

扇形角2α，其面积169α；中心三角形底24、高5，面积60，作差。

M1	扇形面积
M1	由底高求三角形
A1	正确弓形面积

(c) · 2 marks

$26\arccos(5/13) + 24 \approx 54.6$ cm.

边界由横杆24和角2α、半径13的弧组成，代弧度角后取3位有效数字。

M1	弧加弦
A1	数值与精度

Question 17

P1-C07-R17 · 7 marks

(a) · 3 marks

$\dfrac{r^2/2}{\pi r^2/4} = 2/\pi$.

三角形的两直角边都是r，面积$r^2/2$；四分之一圆面积$\pi r^2/4$，约去r^2说明与大小无关。

M1	两部分面积正确
M1	用整个四分之一圆作分母
A1	比例与独立性

(b) · 2 marks

$9\pi - 18$ cm^2.

$r^2/2=18$给$r^2=36$；扇形面积9π，扣18。

M1	由三角形面积求r平方
A1	精确弓形面积

(c) · 2 marks

Impossible: the fraction is fixed at $2/\pi \approx 63.7\%$ for every $r > 0$.

改变半径让两面积同倍增长，不改变占比；63.7%与70%不相等。

B1	固定比例或同倍变化理由
B1	正确设计判断

Question 18

P1-C07-R18 · 6 marks

(a) · 3 marks

Stated formulae.

周长$r\theta + 2r = 30$。弦长$2r\sin(\theta/2)$，由$r\theta - 2r\sin(\theta/2) = 2$代入$r$并整理。

M1	周长表达式
M1	弦长及差值方程
A1	整理正确

(b) A trial angle is $\theta = 1.5$. Using the perimeter constraint,

(i) · 2 marks

1.17 cm.

$r = 30/3.5 = 60/7$，差值$(60/7)[1.5 - 2\sin 0.75] = 1.1719\ldots$，即$1.17$ cm。

M1	同时更新半径并算弧弦差
A1	数值

(ii) · 1 marks

No; the difference is less than 2 cm.

试算差$1.17 < 2$，故不符合要求。这里只检验给定试值，不要求求解三角方程。

B1	依据计算比较

Question 19

P1-C07-R19 · 7 marks

(a) · 2 marks

$3\pi/5$.

每分钟转$2\pi/60$，18分钟转$18(2\pi/60)=3\pi/5$。

M1	时间比例化为弧度
A1	精确角

(b) Between 12:00 and 12:18, find

(i) · 3 marks

$36\pi/5$ cm; 19.4 cm.

走过弧长$12(3\pi/5)=36\pi/5$；位移是弦$24\sin(3\pi/10)=19.416\ldots$。

M1	弧长
M1	弦长
A1	两个结果与单位

(ii) · 2 marks

$216\pi/5\ \text{cm}^2$.

扇形面积$\tfrac12(12)^2(3\pi/5)=216\pi/5$。

M1	弧度制扇形公式
A1	精确面积

Question 20

P1-C07-R20 · 8 marks

(a) · 3 marks

Radius 10 m; angles $0.8, 1.2$ radians.

总弧长20=2r，故r=10，再各以弧长除以r求角。

M1	两相邻弧合成长角对应弧
A1	半径
A1	两角

(b) · 2 marks

$20 - 20\sin 1 \approx 3.17$ m.

新路是AC弦，长$2r\sin(2/2)=20\sin1$；从原20m路线减去。

M1	半角求AC弦
A1	节省长度及精度

(c) · 3 marks

$100 - 50\sin 2 \approx 54.5$ m^2.

两路线间的带状弓形面积为角2、半径10的扇形100，减中心三角形$50\sin2$。

M1	路线间区域识别为弓形
M1	扇形减三角形
A1	面积与单位

Question 21

P1-C07-R21 · 9 marks

(a) · 4 marks

$\theta \le 2\pi - 24/5$ and $\theta \ge 2\pi - 5$.

剩余角$2\pi-\theta$，面积$25(2\pi-\theta)/2 \geqslant 60$；周长$5(2\pi-\theta)+10 \leqslant 35$，分别解。

M1	面积不等式
A1	面积约束
M1	周长含两半径
A1	周长约束

(b) · 2 marks

$2\pi - 5 \le \theta \le 2\pi - 24/5$.

两约束取交集，得到闭区间；两端约1.283和1.483，均在原域$(0,\pi)$内。

M1	两约束与原范围取交
A1	包含等号的区间

(c) · 3 marks

$10\sin\left((2\pi - 5)/2\right) \approx 5.98$ cm.

缺口弦长$10\sin(\theta/2)$，在$0<\theta<\pi$内随角增大；取允许的最小角$2\pi-5$。

M1	弦长公式
M1	在题设角范围选择最小θ
A1	数值及单位

Question 22

P1-C07-R22 · 8 marks

(a) · 3 marks

$r = 8$ cm; $\theta = 2\arccos(8/17)$.

半弦15、斜边17，垂距$\sqrt{(289-225)}=8$；由辅助直角三角形$\cos(\theta/2)=8/17$。

M1	勾股求垂距
A1	内半径
A1	共角表达式

(b) · 2 marks

$225\theta/2$ cm^2.

面积$\tfrac{1}{2}(17^2-8^2)\theta = 225\theta/2$。弦AB不是要从此区域中扣去的边界。

M1	两扇形面积之差
A1	正确化简

(c) · 3 marks

$25\theta + 18 \approx 72.0$ cm.

内外弧共$(17+8)\theta$，两径向边各$17-8=9$，总长$25\theta+18$。

M1	两曲边合计
M1	两直边
A1	周长及精度

Question 23

P1-C07-R23 · 8 marks

(a) · 3 marks

$r = 4$ cm; $\theta = 4.5$ radians.

面积可写½r×弧长，所以36=9r给4；18=rθ给θ4.5。

M1	用面积与弧长消去θ
A1	半径
A1	角度

(b) · 3 marks

Major, since $4.5 > \pi$; perimeter 26 cm.

θ超过π而小于2π，故为大扇形；周长为18加两半径8。

B1	大扇形及角度依据
M1	完整周长
A1	数值

(c) · 2 marks

$16\pi - 36$ cm^2.

另一扇形角2π−4.5，或从圆面积16π中减36；不能把同一弧角重复使用。

M1	用互补整圆角或面积
A1	精确面积

Question 24

P1-C07-R24 · 8 marks

(a) · 3 marks

From $3r + 3r\theta = 54$ and $\frac{1}{2}r^2(3\theta) = 108$, eliminate θ to obtain the equation.

围栏含三条径向边，总长3r，曲边总长3rθ。由面积得3rθ=216/r，代总长式乘r并除3。

M1	三条径向围栏及曲边关系
M1	面积关系并消元
A1	推出二次式

(b) · 3 marks

$r = 12$ m; $\theta = 0.5$, $2\theta = 1$ radian. Reject $r = 6$, which gives total angle $6 > \pi$.

二次式分解为(r−6)(r−12)。总角216/r²；r=6给6弧度，违背总角小于π；r=12给1.5，故分角0.5、1。

M1	两个候选半径
M1	检查总角限制
A1	正确半径和两角

(c) · 2 marks

42 m.

从总围栏54减一条内部半径12，得42；也可算曲边18加两外半径24。

M1	排除内部围栏
A1	外周长度

Question 25

P1-C07-R25 · 9 marks

(a) · 3 marks

Height 8 cm; central angle $2\arcsin(3/5)$.

半宽6、斜边10，勾股得高8；以垂直对称线二等分中心三角形，半角正弦6/10=3/5。

M1	辅助直角三角形
A1	高度
A1	夹角表达式

(b) Let $\alpha = \arcsin(3/5)$.

(i) · 3 marks

$100\alpha - 48$ cm^2.

扇形角2α给面积100α；中心三角形底12、高8，面积48，相减。

M1	正确扇形面积
M1	正确中心三角形面积
A1	弓形精确表达式

(ii) · 3 marks

Sides $50\pi - 48 - 100\alpha \approx 44.7$ cm^2; cap 16.4 cm^2; sides are larger.

总空余面积50π−96；扣帽形后两侧为50π−48−100α。代α约0.643501比较，不可将整个空余区当作帽形。

M1	半圆减矩形再减帽形
A1	两个数值
B1	比较结论

Question 26

P1-C07-R26 · 8 marks

(a) · 3 marks

$\theta = 3, r = 6$.

rθ=3r且r>0给θ=3；面积$3r^2/2=54$，取正根6。

M1	弧长比例条件
M1	面积求正半径
A1	两值

(b) A smaller concentric sector with the same angle is removed, leaving an annular sector of area $30\ \text{cm}^2$. Find

(i) · 3 marks

4 cm.

去除面积54−30=24，因此$3s^2/2=24$，s=4。

M1	正确去除面积
M1	扇形关系
A1	内半径

(ii) · 2 marks

34 cm.

两弧长3(6+4)=30，两径向边各2，合34。

M1	计入两弧和两直边
A1	周长

Question 27

P1-C07-R27 · 10 marks

(a) · 4 marks

Distance 8 cm; angle $2\arcsin(3/5) \approx 1.29$ radians.

半弦6与半径10构成直角三角形；勾股得中心距8，半角正弦3/5。

M1	半弦直角三角形
A1	距离
M1	半角关系
A1	弧度角

(b) For the remaining larger piece, find

(i) · 3 marks

$100\pi - 100\alpha + 48$ cm^2.

原圆100π减小弓形(100α−48)。

M1	小弓形面积
M1	整圆减去小弓形
A1	精确面积

(ii) · 3 marks

$20\pi - 20\alpha + 12$ cm.

保留大弧角2π−2α，弧长10倍，再加弦12。

M1	使用大弧而非小弧
M1	加直弦
A1	周长

Question 28

P1-C07-R28 · 7 marks

(a) · 3 marks

$2.4(1-\cos\phi)=0.6;\ \phi=\arccos(3/4)\approx 0.723.$

端点竖直投影为2.4cosφ，升高是长度减去投影；按指定角度范围取唯一角。

M1	高度投影关系
A1	精确余弦
A1	弧度值

(b) During one complete swing from the left-hand extreme to the right-hand extreme, find, to 3 significant figures,

(i) · 2 marks

$4.8\arccos(3/4)\approx 3.47$ m.

转过总角2φ，弧长2.4×2φ。

M1	使用总角2phi
A1	路程

(ii) · 2 marks

$1.2\sqrt{7}\approx 3.17$ m.

两端对称，弦长2×2.4sinφ；sinφ=√7/4。

M1	弦长或对称投影
A1	距离

Question 29

P1-C07-R29 · 8 marks

(a) · 4 marks

Area $(180-25\pi)\ \text{cm}^2$; perimeter $(36+10\pi)\ \text{cm}$.

两半圆合为整圆去除，面积减25π；两条短直边由两半圆弧替代，保留两条长边36。

M1	两半圆面积合并
A1	面积
M1	正确识别边界
A1	周长

(b) Each semicircular cut is replaced by a sector with the same radius and with its arc length reduced by 20%. For this comparison, consider one cut only. Find

(i) · 2 marks

$4\pi/5$ radians.

原半圆弧角π，固定半径下弧长和角成比例，取0.8π。

M1	固定半径弧角比例
A1	角度

(ii) · 2 marks

$5\pi/2\ \text{cm}^2$.

一半圆面积25π/2；同半径角减少20%，面积减少其20%=5π/2。

M1	固定半径面积比例
A1	面积变化

Question 30

P1-C07-R30 · 8 marks

(a) · 3 marks

$r\theta = 16$, $2r\sin(\theta/2) = 8\sqrt{3}$; divide to obtain the result.

分别列弧长与弦长关系，消去r；不要求用超出阶段范围的方法求这个超越方程。

M1	弧长关系
M1	弦长关系
A1	消去半径目标式

(b) A designer proposes $\theta = \pi/3$.

(i) · 2 marks

$r = 48/\pi$ cm.

仅使用rθ=16并代入提议角。

M1	代弧长式
A1	半径

(ii) · 3 marks

No; proposed chord $48/\pi \approx 15.3$ cm, required $8\sqrt{3} \approx 13.9$ cm.

θ/2=π/6时2r sin(π/6)=r，分别比较未舍入值，明显不等。

M1	计算提议弦长
A1	两比较值
B1	不满足结论